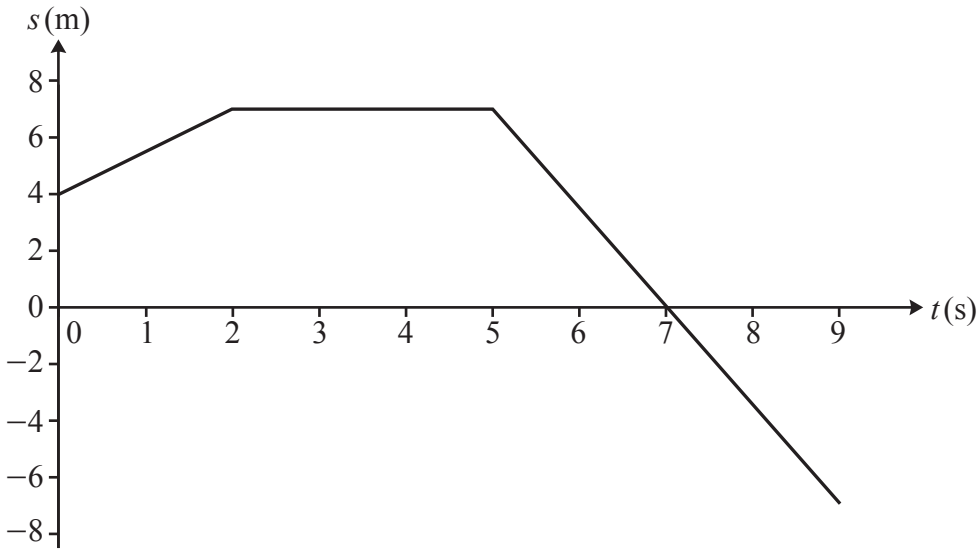
- The acceleration due to gravity is denoted by $g\,\text{m s}^{-2}$. When a numerical value is needed use $g = 9.8$ unless a different value is specified in the question.

5 The graph shows displacement s m against time t s for a model of the motion of a bead moving along a straight wire. The points (0, 4), (2, 7), (5, 7) and (9, –7) are the endpoints of the line segments.



(a) Find an expression for the displacement of the bead for $0 \leqslant t \leqslant 2$. **[2]**

(b) Sketch the velocity-time graph for this model. **[2]**

(c) Explain why the model may not be suitable at $t = 2$ and $t = 5$. **[1]**